Utilizing Low Grade Coal and Oil Sands Residues for Producing Power, Hydrogen and Chemicals in Western Canada-Opportunities and Challenges

IAGT- NRC Gas Turbine Conference
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Ottawa

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Presentation Outline

- Alberta Energy Research Institute- Overview
- Western Canadian Energy Resources
- Oil sands challenges and opportunities
- The CO_2 issue
- Gasification- Key technology platform for oil sands, coal and clean power generation
- IGCC with CO_2 capture
- RD&D needs
- Conclusions

Alberta Energy Research Institute (AERI)
(Energy Technology Arm of the Alberta Government)

www.aeri.ab.ca

Mission: Enhance the development of clean energy resources through research, technology and innovation.

Mandate:

- Position Alberta for the future in energy
- Add value to Alberta's energy resources

Strategy:

- Invest in technology development projects with industry
- Partnerships & International collaboration
- Focus on Technology Platforms: e.g. Gasification, Upgrading and Carbon Capture and Storage (CCS)

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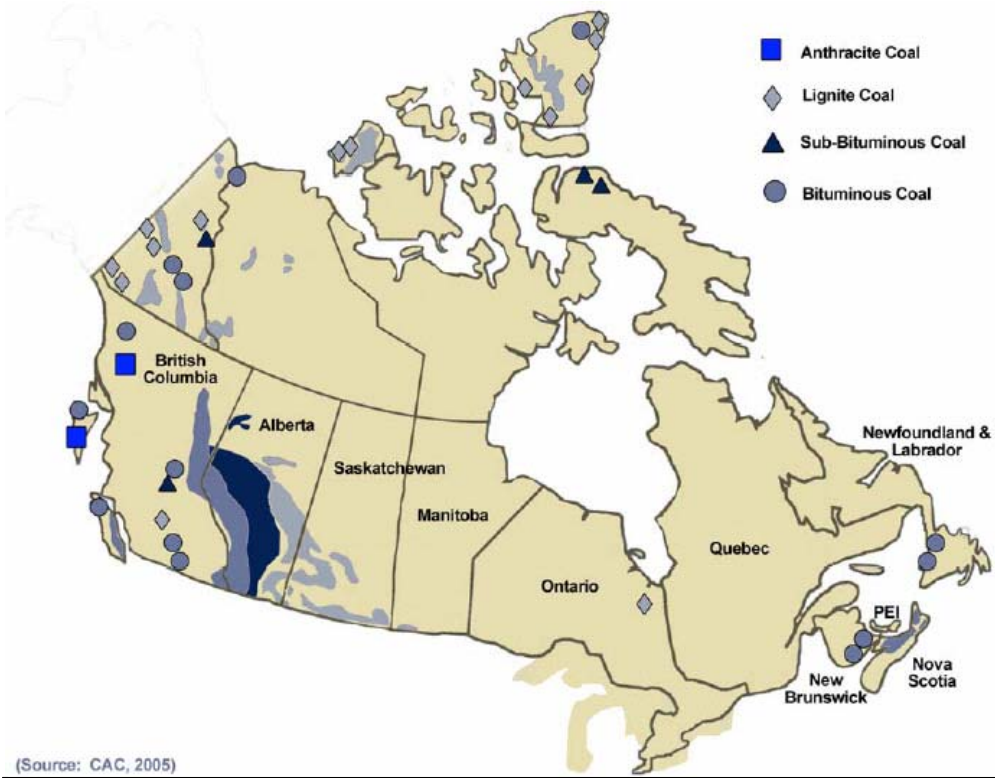
Fossil Reserves Concentrated in Western Canadian Sedimentary Basin

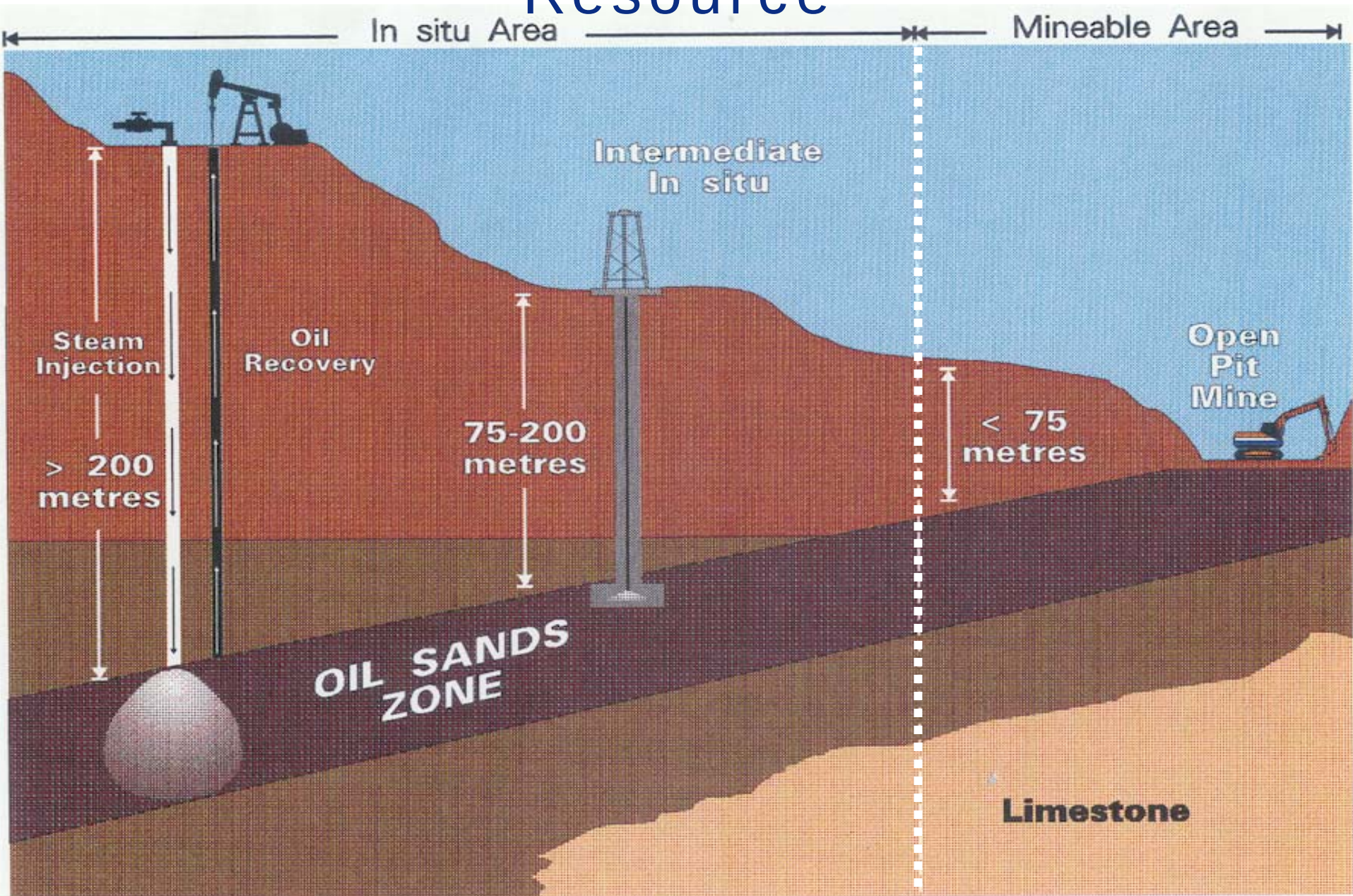


(Source: CAC, 2005)

- Proven oil reserves: 2nd after Saudi Arabia
- Crude oil producer: 7th in the world
- Energy producer: 5th in the world
- Natural gas producer:3rd in the world
- Largest exporter to USA

	Oil 10^9 bbls	Gas 10^{12} cu.ft.	Bitumen 10^9 bbls	Coal 10^9 tonnes
Remaining Reserves	1.5	39	173	34
Annual Production	0.19	4.8	0.48	0.037
Reserves/Production, years	9.4	8.1	360	919

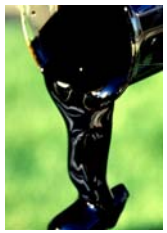
The Nature of the Oil Sands Resource



The Bitumen Value Chain- Product and Market Options

	Wt %
API	7 - 11.0
S	4.6 – 5.2
N	0.37 – 4.5
O	0.3 – 0.5
H	10 - 10.8

	Wppm
Nickel	68-82
Vanadium	190-210



Bitumen

Oil Sands
10% Bitumen

Integrated Processing
upgrading, refining & conversion to petrochemicals

Difficult to process and energy intensity is high

SCO

Fuel Products
Synthetic Crude Oil
Diesel
Kerosene
Gasoline
SNG (methane)

Petrochemicals
Naphtha
Gasoil
Ethylene
Propylene
Butadiene
Hydrogen
Aromatics

Chemicals & Fertilizers
Syngas/Hydrogen
Methanol
Ammonia
Ammonium Nitrate
Sulphur
Urea
Urea Ammonium Nitrate

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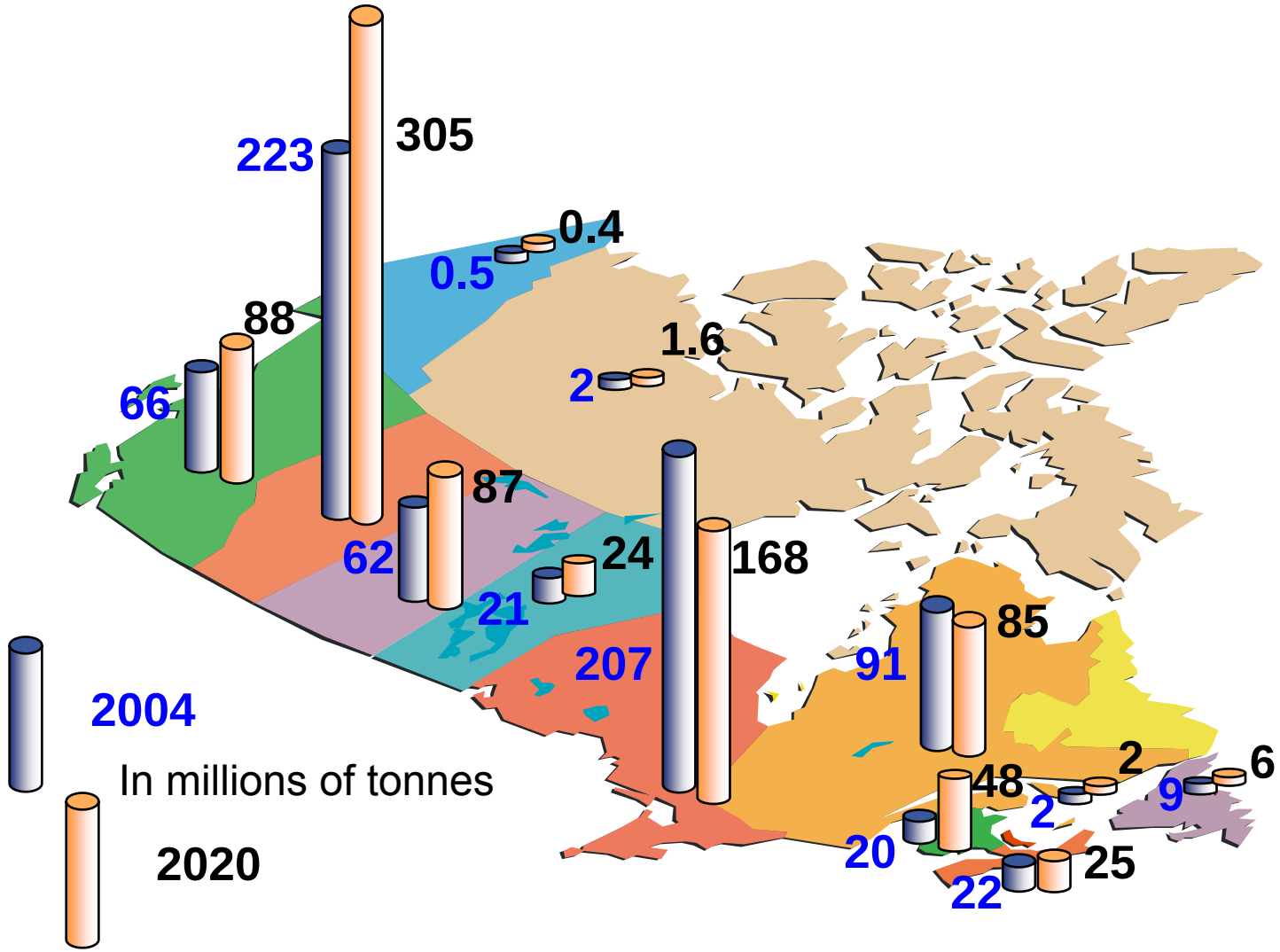
Alberta is responsible for approximately 33% of Canada's GHG emissions (2006)



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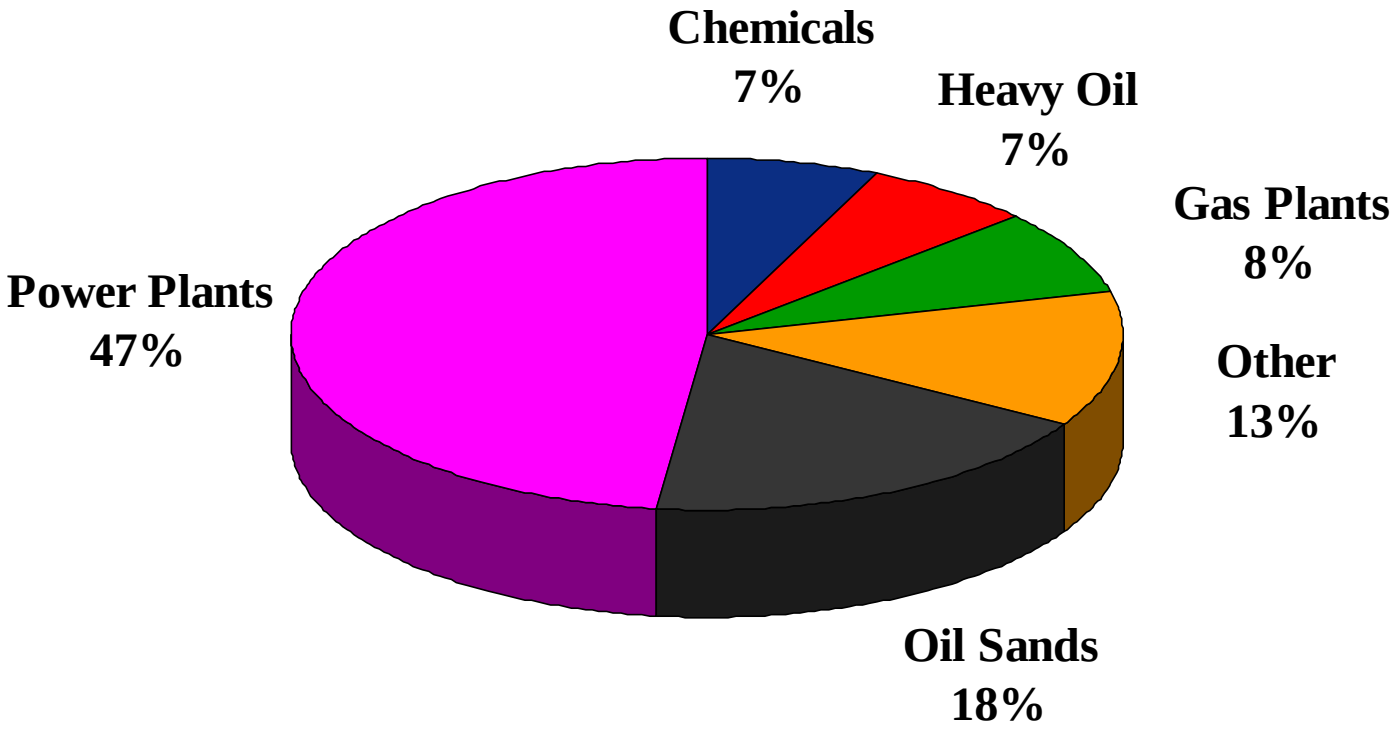
Large Industrial Emitters Profile

(>100,000 tonnes CO2e/year)

50% of emissions come from large point-sources



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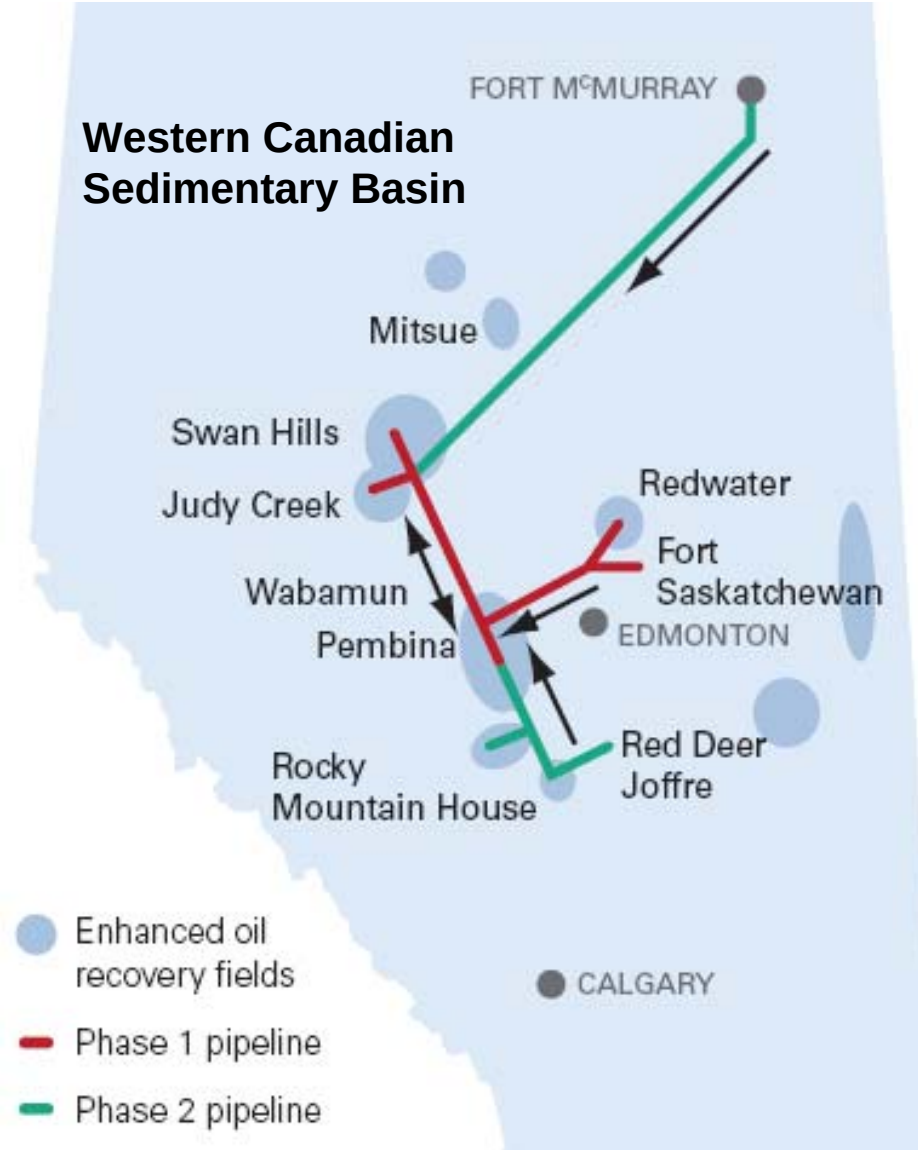
Good News: Emitters are close to Oil Fields and Deep Saline Acquifers

Integrated CO_2 Network(ICO_2N)
Alliance of 15 of Canada's large industrial companies, representing

- over 100Mt CO_2 of annual emissions
- 60% of Alberta's power generation,
- 95% of the oil sands producers
- Connect large CO_2 emitters to EOR and sequestration sites

"WCSB has vast potential for CCS in deep saline acquifers and oil fields suitable for enhanced oil recovery"
Summary Report by the ICO_2N Group of Companies - December 2007.



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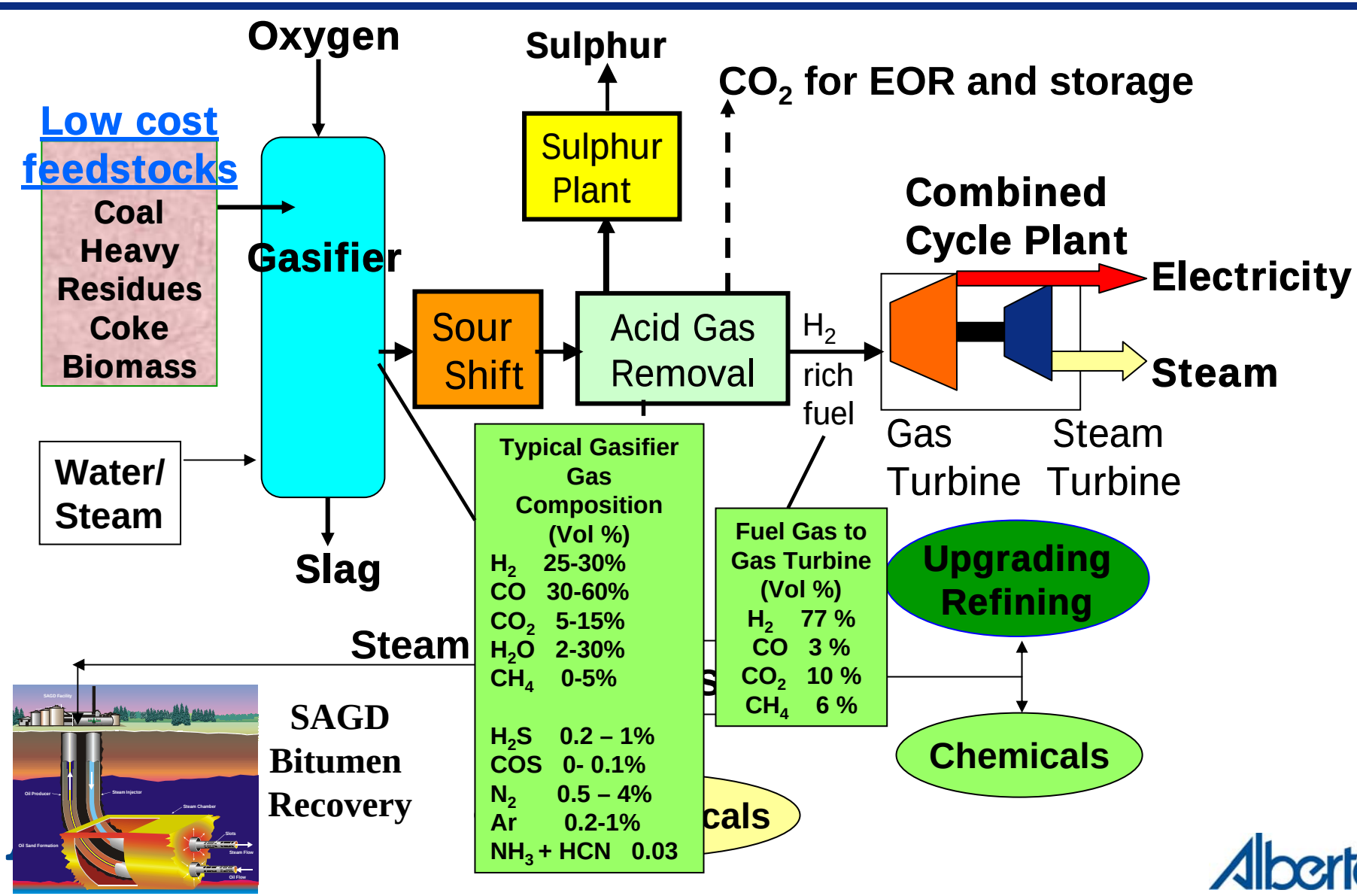
Alberta's Approach to Reduction of GHG's

- Long term issue
  - Need to start with practical, achievable objectives
- Technology will be a big part of the long-term solution
- Introduced multi-sector regulatory-based demand for carbon reductions
  - Climate Change and Emissions Management Act
  - Ecotrust Fund
- Announced $2 billion fund for CCS projects capturing 5 million tonnes per year of CO2
  - Announced June 2008
  - Strong response to request for expressions of interest

Gasification
Drivers, Opportunities & Challenges

Drivers and Opportunities	Challenges
• Large coal resources • Electricity demand growth • Aging coal plants • Oil sands developments • Natural gas replacement • GHG regulations • Proximity of EOR and sequestration sites • Gasification of coal & oil sands residues offers sustainable solutions, BUT face significant challenges	• Capital cost • Reliability • CO_2 & water management • Performance with low quality fuels

Gasification: A Key Technology Platform



Gasification Feedstocks Options

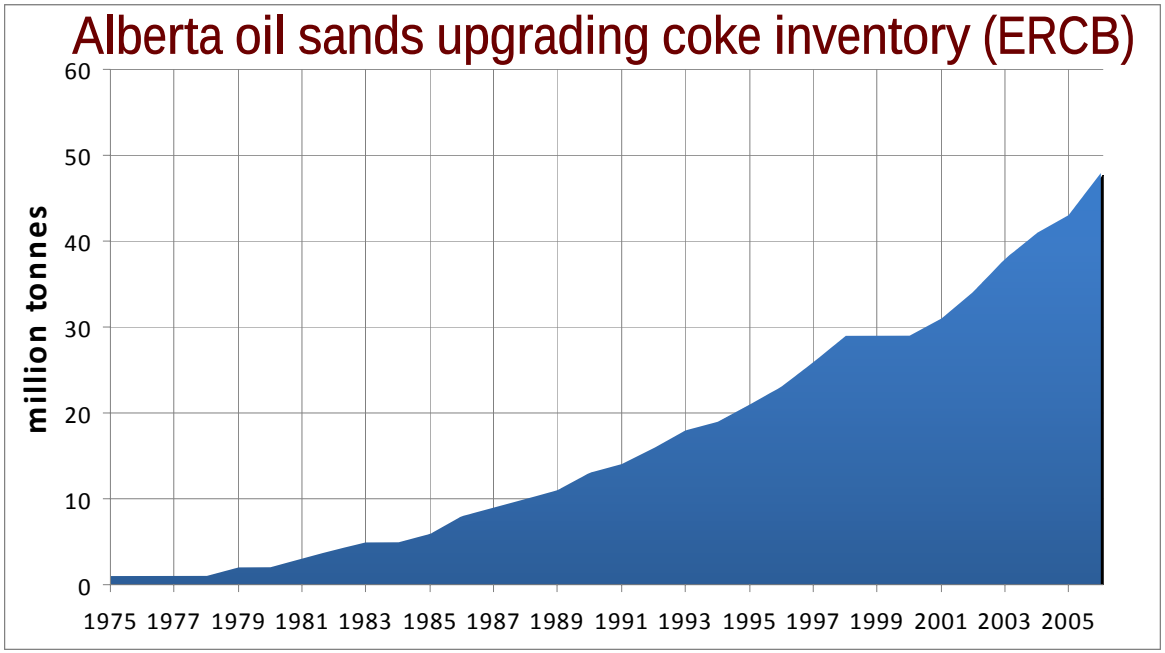
	Coal		Oil Sands Residues			
	LIGNITE	SUB-BIT	DEL. COKE	FLUID COKE	ASP'TENE	
COAL PROPERTIES						
MOISTURE (AR)	33.54 (1)	20.00	10.07 (2)	0.24 (2)	0.22 (2)	wt%
ASH (AR)	13.46	13.93	7.00	7.00	7.00	wt%
Carbon	59.55	61.11	90.27	89.03	84.07	wt%
Hydrogen	3.87	3.53	3.75	1.88	7.50	wt%
Oxygen	14.59	16.88	0.00	0.00	0.00	wt%
Sulphur	0.74	0.33	4.16	6.93	7.23	wt%
Nitrogen	1.01	0.75	1.82	2.16	1.20	wt%
HHV	14963	18930	36208	33705	38700	kJ/kg

(1) High sodium coal

(2) High Ni and Va

Oil Sands Coke- Underutilized Energy Reserve for Producing Hydrogen and Power

- **Stockpiles of petcoke (2007) ~ 54 million tonnes**
- **Coke production ~ 20,000 TPD; Future (2015): ~ 40,000 TPD**



Gasification of coke can replace natural gas as a source of hydrogen, power and heat for oil sands operations, and form the basis for a world scale chemical and petrochemical cluster

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Joint AERI-Industry Gasification Projects

- Gasification of oil sands residues to replace natural gas as a source of hydrogen and fuel gas
  - CO2 capture from OPTI Integrated SAGD-Gasification operation
- Coal-Based electric power generation with CO_2 capture
  - Canadian Clean Power Coalition (CCPC) Industry/AERI/EPRI
  - EPCOR/CCPC/AERI/Fed.Gov. FEED study
- Joint AERI-Industry program to develop advanced 'clean' technologies
- Chemical cluster study based on synthesis gas from gasifying oil sands coke

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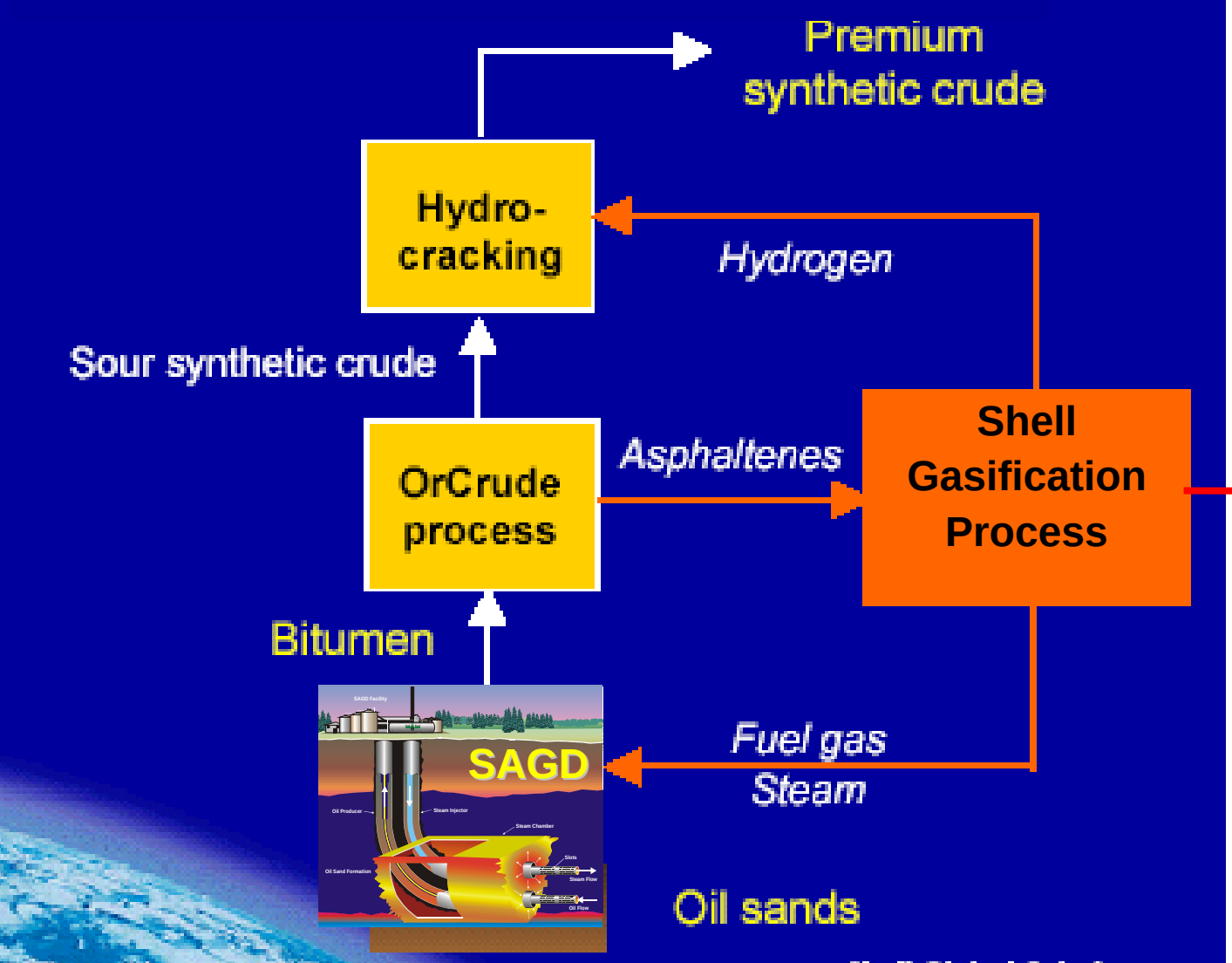
Canada's First Commercial Gasification Plant - Integrated with SAGD

Opti – Nexen Long Lake Project



Joint Opti-AERI Project

Optimum CO_2 capture schemes for existing and future phases

CO2 Capture

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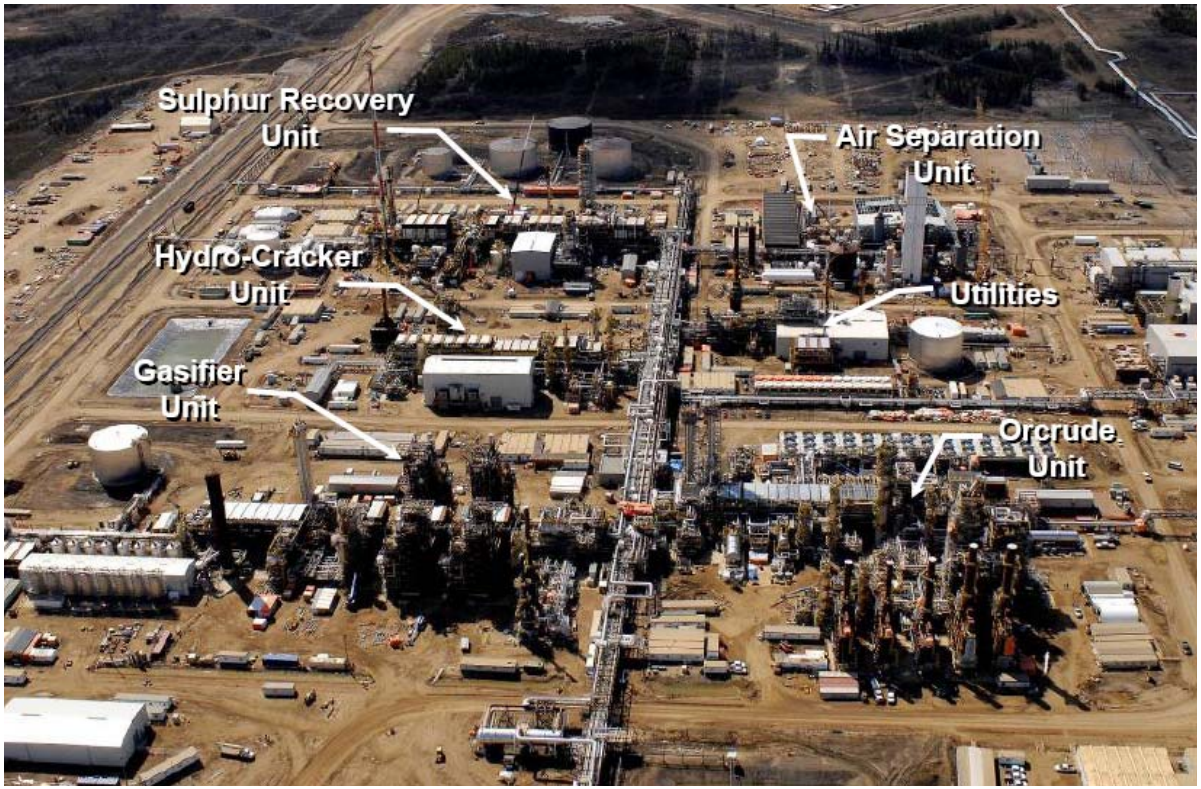
Opti Long Lake Plant



Long Lake Upgrader Facility (April 2007

- One of four gasifier trains
- Largest Shell liquid fed gasifier trains in the world (when designed)

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The Canadian Clean Power Coalition (CCPC)

- An association of Canadian and U.S. coal and coal-fired electricity producers including EPRI
- Goal: Demonstrate that coal-based electricity generation can effectively address all environmental issues projected in the future, **including CO_2**
- Provide flexible fuel capability– bituminous, sub-bituminous, lignite, and petroleum coke

www.canadiancleanpowercoalition.com

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CCPC Technologies Considered & Evaluated

Gasification Technologies

Phase 1- High & low rank coals

- GE Energy
- Shell

Performance & Cost estimates

- ConocoPhillips
- British Gas Lurgi
- EAGLE
- High Temperature Winkler
- Sasol-Lurgi
- KBR Transport Gasifier

Phase 2- Low rank coals

- Advanced slurry feed gasifier
- Advanced dry feed gasifier

CO_2 Capture Technologies

- Gasification (IGCC)
  - Sour shift
  - Gas cleanup
  - H_2 rich fuel
  - GE 7FB Frame Turbine
  - 90 % CO_2 Capture
- Amine scrubbing
- Oxyfuel combustion

- ***Other emission levels equivalent to NGCC plants***

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Impact of CO_2 Capture on IGCC Performance and Cost

	Lignite			Subbituminous Coal		
	Non-Capture	Capture	Change %	Non-Capture	Capture	Change %
Net Plant Output, MW	571	483	- 25.4	524	481	- 8.2
Plant Efficiency, % (HHV)	40.9	31.5	- 9.4	39.6	32.6	- 8.0
Capital Cost, $/kW Net	5480	7186	+ 31	4521	5547	+ 27
CO2 emission intensity (t/MWh)	0.86	0.18	- 79	0.87	0.27	- 70

- IGCC performance and costs depends on fuel quality
- CO_2 capture reduces plant performance and increases plant costs
- CO_2 emission intensity reduced 70+%

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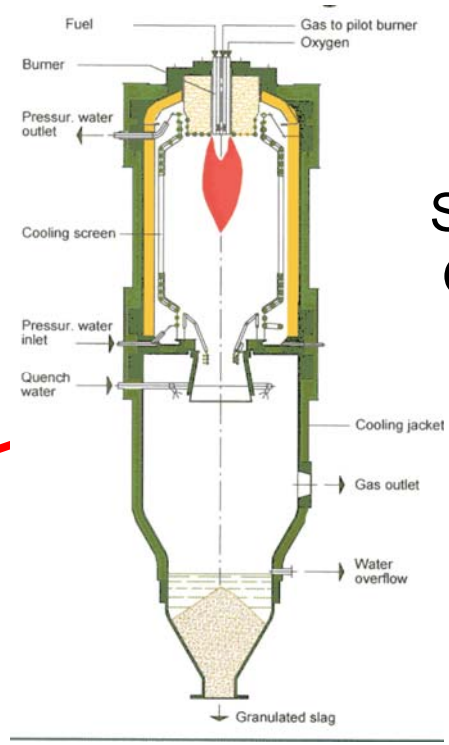
Conclusions
IGCC with CO_2 Capture

- Technology enhancements needed to make IGCC more competitive especially for low rank western coals.
- IGCC with CO_2 capture needs to be demonstrated at full scale to prove the designs and optimize performance
- Site specific FEED studies are needed to improve reliability of cost estimates.

EPCOR FEED Study- IGCC with CO_2 Capture



Siemens Gasifier

Courtesy EPCOR

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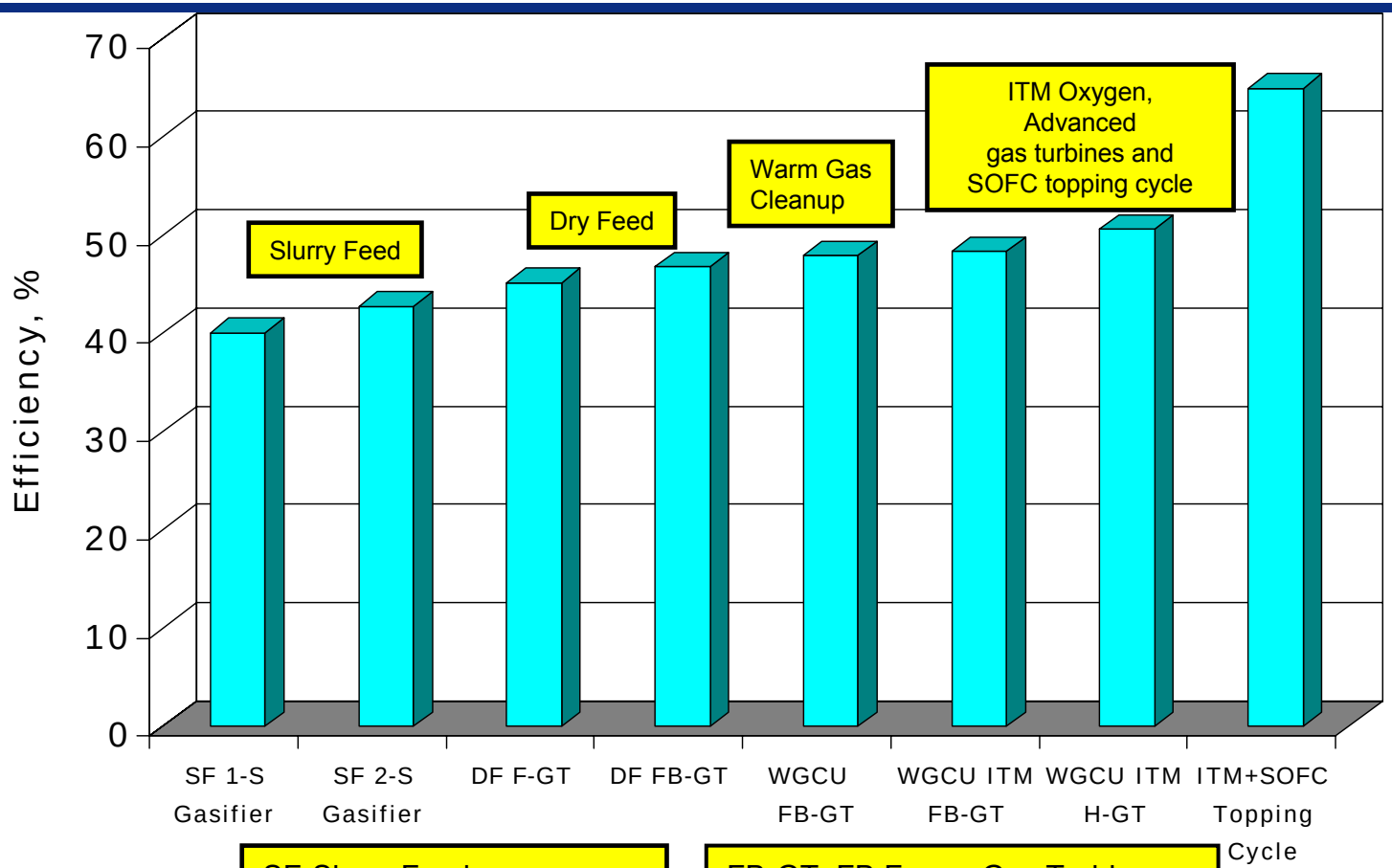
RD&D to Improve IGCC Performance

- Improve feedstock quality. Reduce ash and moisture
- High pressure dry solids feed pumps
- Advanced gasifiers
- Warm gas clean-up
- Ion Transport Membranes for oxygen generation
- Hydrogen transport membranes
- Advanced gas turbines
- Integrated SOFC-Gas Turbine system
- Enhanced/new CO_2 capture technologies

IGCC Without CO_2 Capture: Potential Benefits of RD&D



SF-Slurry Feed
1-S Single Stage
DF- Dry Feed
F-GT: F-Frame Gas Turbine

FB-GT: FB Frame Gas Turbine
WGCU- Warm Gas Cleanup
ITM- Oxygen Transport Membrane
SOFC- Solid Oxide Fuel Cell

Adapted from: The Potential Impact of Developing Technologies on the Economics and Performance of Future IGCC Plants, John Plunkett, David Gray, Charles White, Noblis, USA International Pittsburgh Coal Conference, Sept 2008

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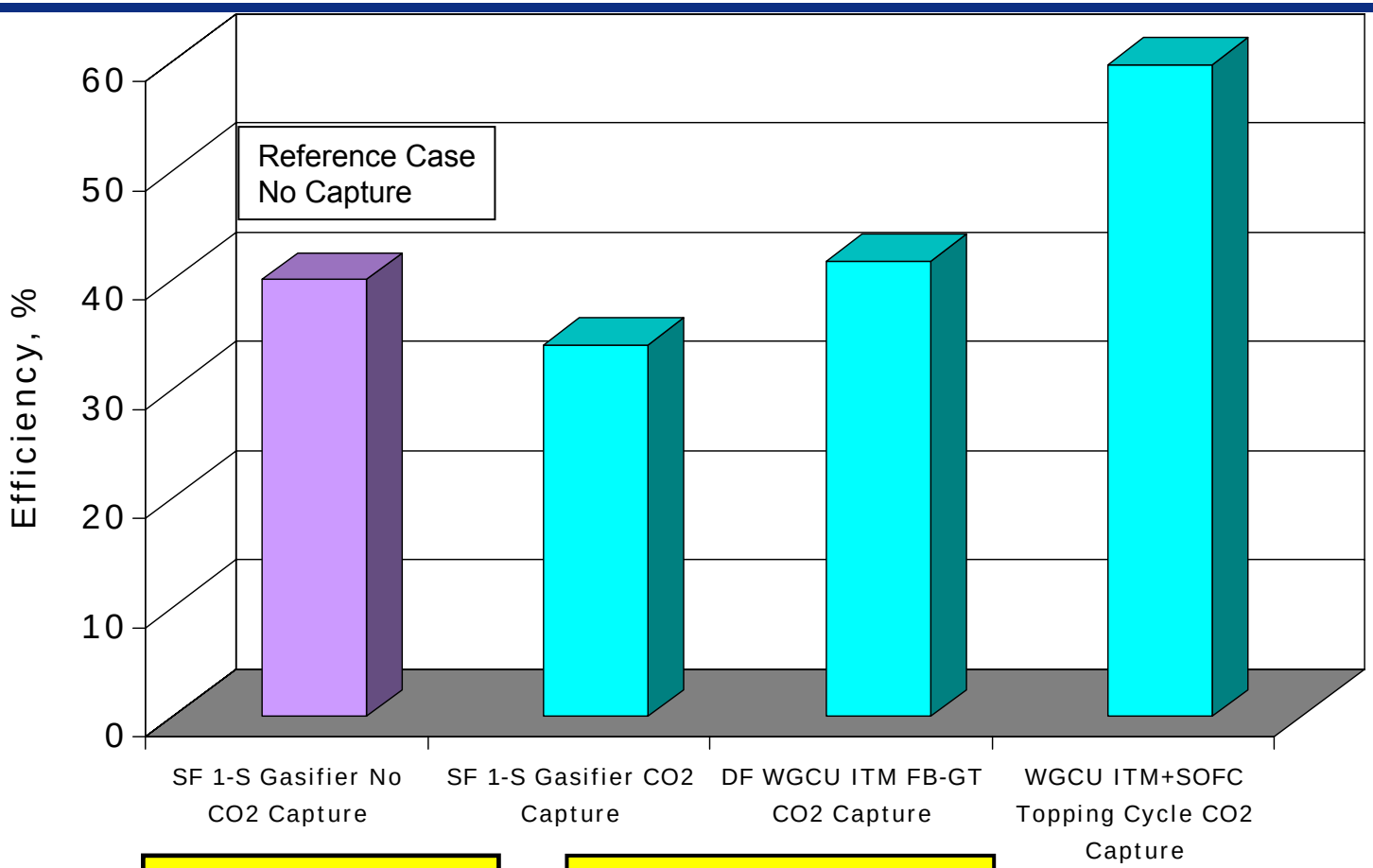
IGCC With CO_2 Capture Potential Benefits of RD&D



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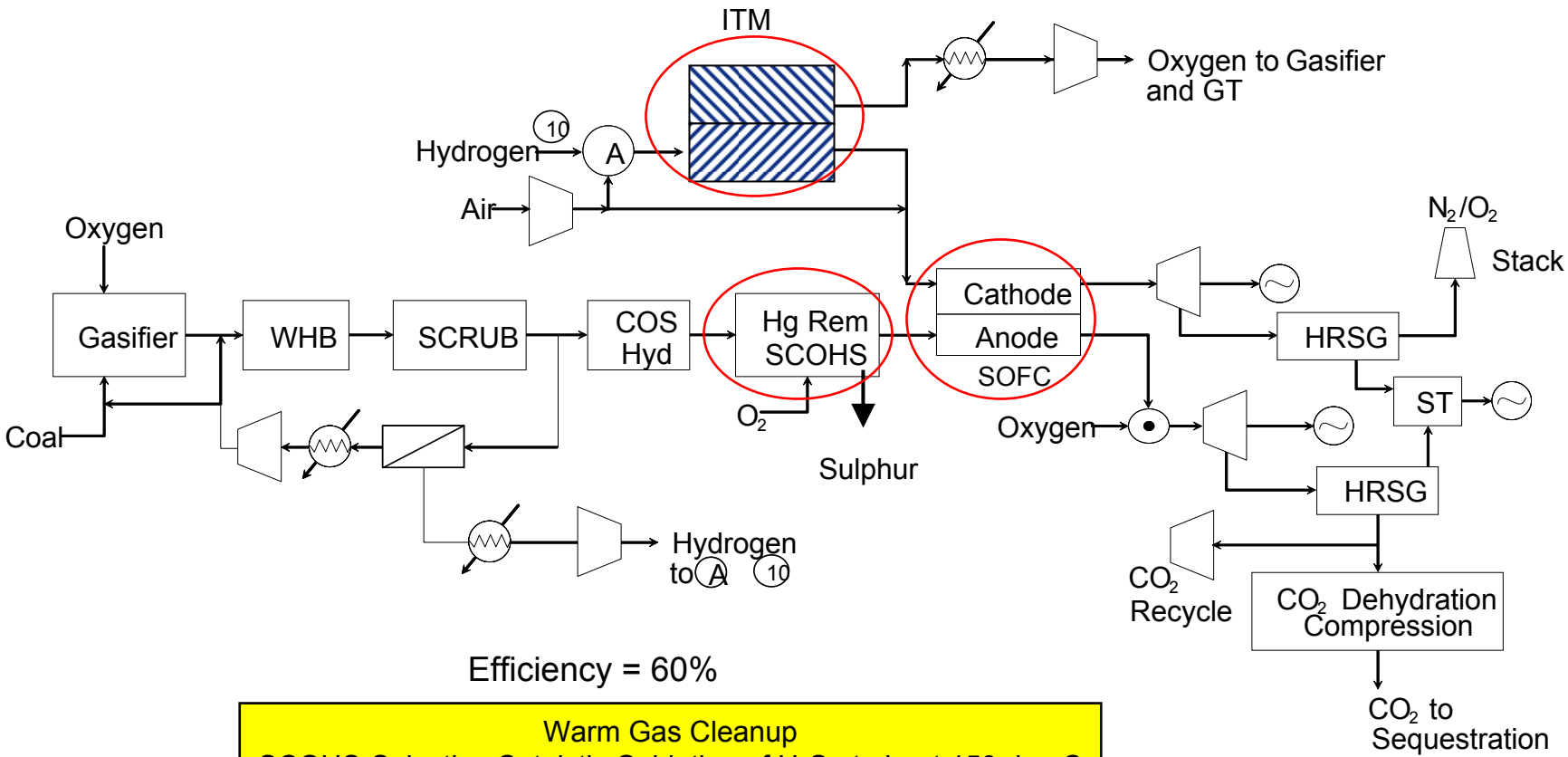
IGCC with ITM, Warm Gas Cleanup, SOFC topping cycle and CO_2 Capture



Efficiency = 60%

Warm Gas Cleanup
SCOHS-Selective Catalytic Oxidation of H_2S at about 150 deg C
H2S + 1/2O2 = S + H2O
Eliminates the need for sulphur plant

Adapted from: The Potential Impact of Developing Technologies on the Economics and Performance of Future IGCC Plants, John Plunkett, David Gray, Charles White, Noblis, USA International Pittsburgh Coal Conference, Sept 2008

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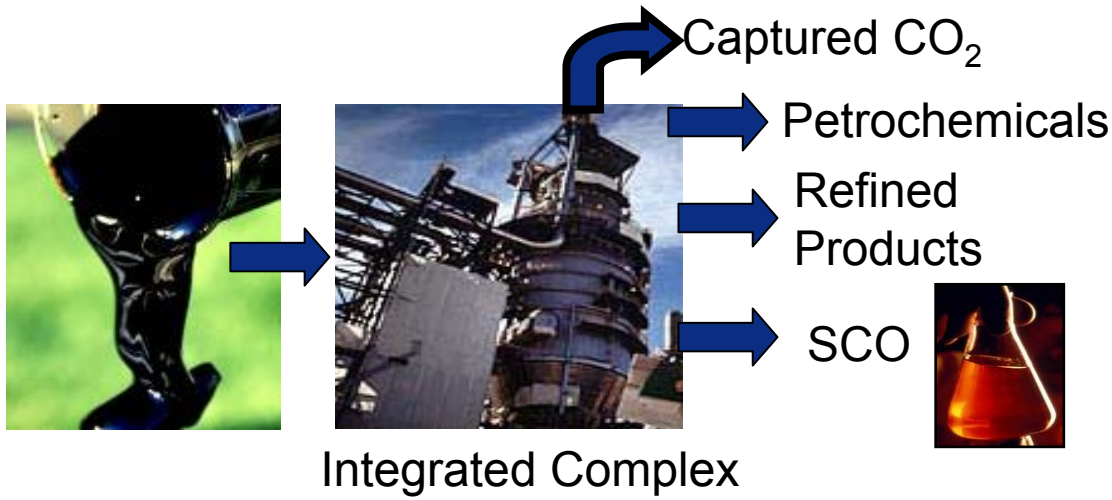
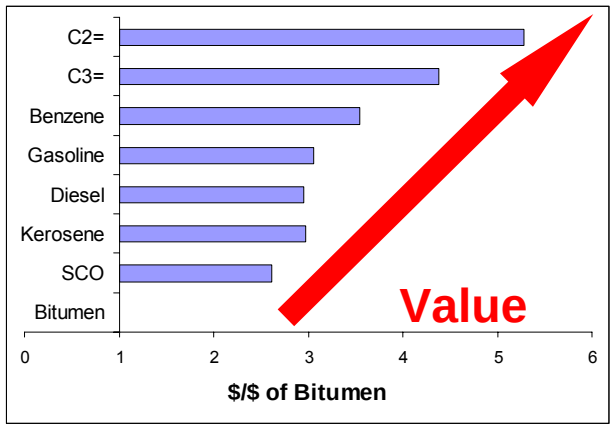
AERI-Industry Program to Develop Advanced 'Clean' Upgrading Technologies

- Goal: Develop next generation technologies for upgrading bitumen to high value products more economically and with lower environmental footprint





- Screened 100 technologies: selected eight for further development in partnership with process developers and first adopters
  - Includes 3 gasification technologies

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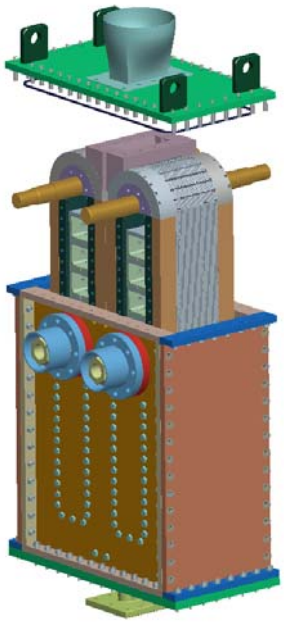
Pratt and Whitney Rocketdyne Technologies

- Rapid Mix Burner
- Cooled Liner
- Spray Quench

High Mass Flux Entrained Bed Gasifier



High Pressure Solids Feed Pump (Conceptual)

Gasifier Pilot Facility at GTI 8/15/08

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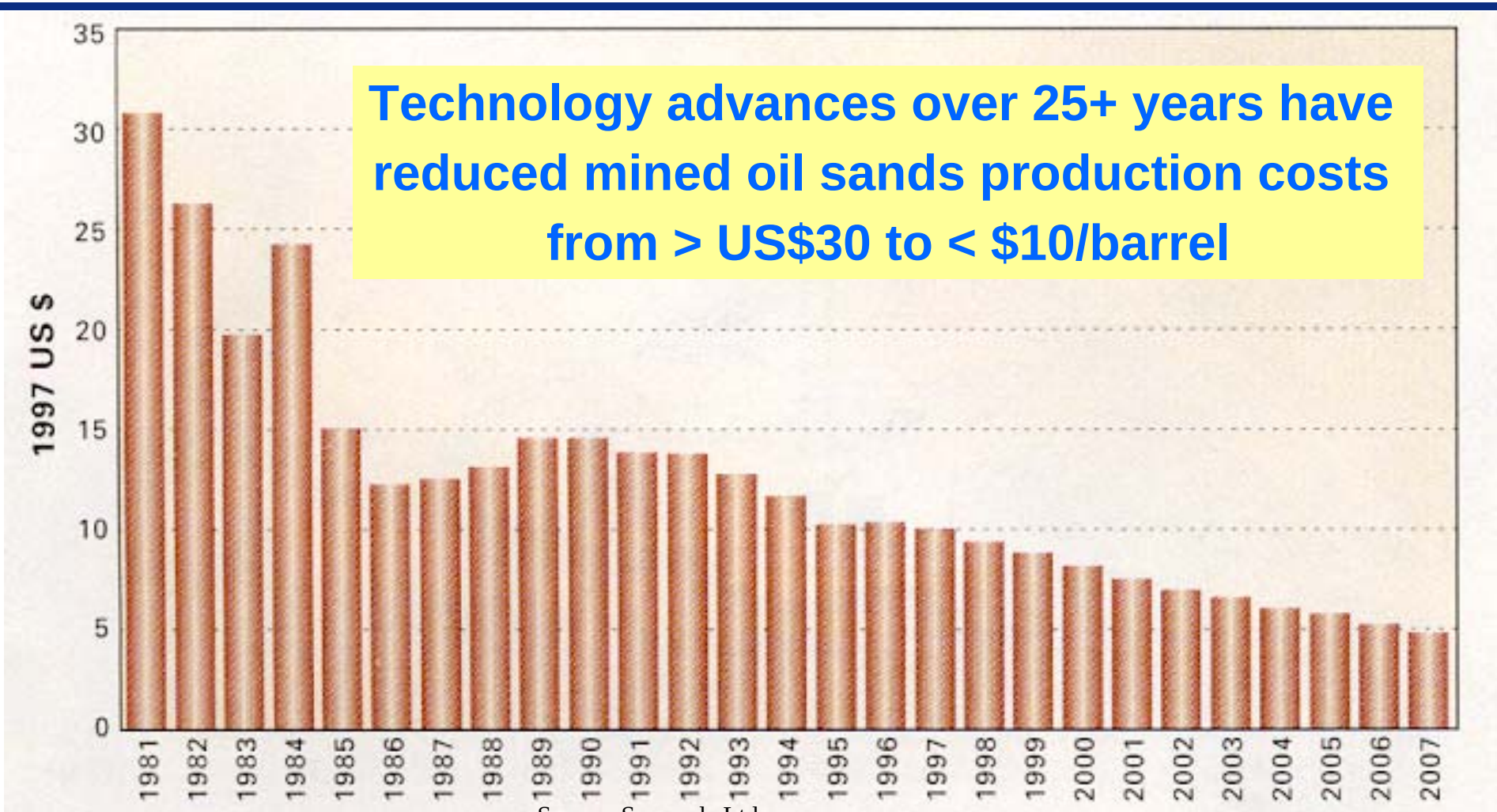
Conclusions

- Driver for gasification and IGCC in Western Canada is conversion of low coal and oil sands residues into clean power, hydrogen and high value products while capturing CO_2
- RD&D to improve overall performance and cost competitiveness include
  - Advanced gasifiers and gas turbines
  - Advanced gas cleanup technologies
  - Membranes for producing oxygen and separating hydrogen & CO_2
  - Advanced CO_2 capture technologies
  - Integration and optimization of all system components.
- First of a kind demonstration plants are needed to accelerate commercial acceptance and deployment.

The Oil Sands Model
"Learning by Doing"



Source: Syncrude Ltd.

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